

DATA VISUALIZATION

installing matplotlib , seaborn , pandas, numpy,

```
!pip install matplotlib seaborn pandas numpy
```

Show hidden output

Implementing matplotlib

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
```

matplotlib → complete visualization library

pyplot → module used for plotting graphs

plt → shortcut name (alias)

```
# Every chart follows this pattern:
# 1. Create figure & axes
# 2. Plot data

# 3. Add labels  4. Show/Save
```

Line Chart — Trends over time

```
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [1200, 1500, 1100, 1800, 2100, 1950]
costs = [800, 900, 850, 1000, 1100, 1050]
```

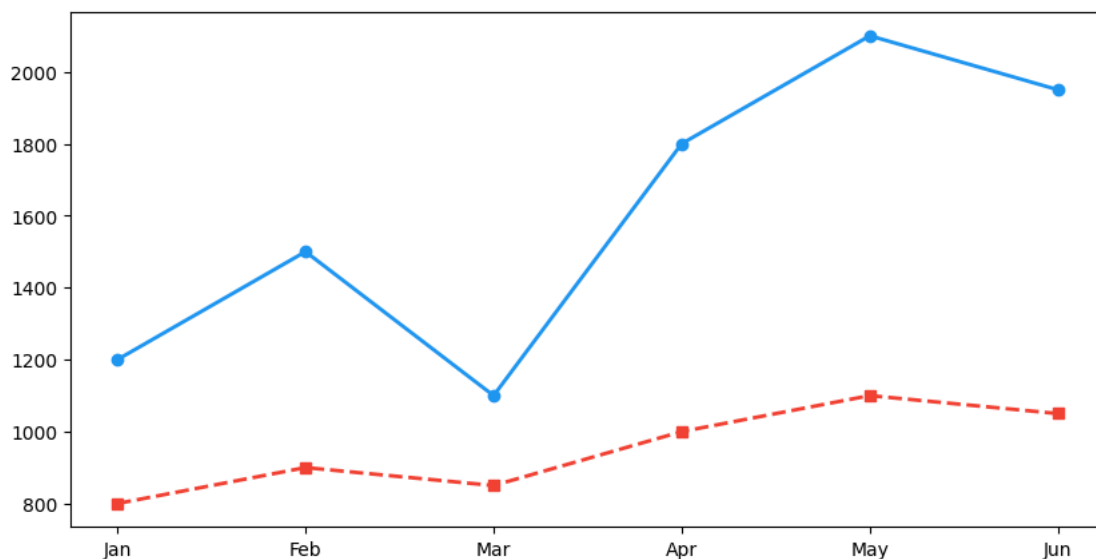
```
plt.figure(figsize=(10, 5))

# width, height in inches
plt.plot(months, sales, marker='o', label='Sales', color='#2196F3', linewidth=2)

#plt.plot is used to draw line graphs
# | Marker | Shape |
# | ----- | ----- |
# | 'o' | circle |
# | 's' | square |
# | '^' | triangle |
# | '*' | star |
# | '+' | plus |
# | 'x' | x shape |

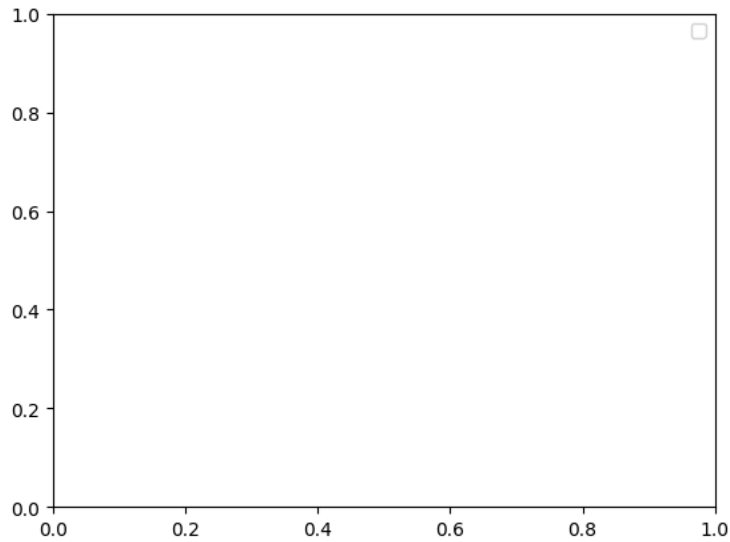
plt.plot(months, costs, marker='s', label='Costs', color='#F44336', linewidth=2, linestyle='--')
```

[<matplotlib.lines.Line2D at 0x7ba328703e30>]



```
plt.legend()
```

```
/tmp/ipykernel_1525/4061938096.py:1: UserWarning: No artists with labels found to put in legend. Note that artists whose labels are not unique are not plotted in legend.  
plt.legend()  
<matplotlib.legend.Legend at 0x7ba328a6b1a0>
```

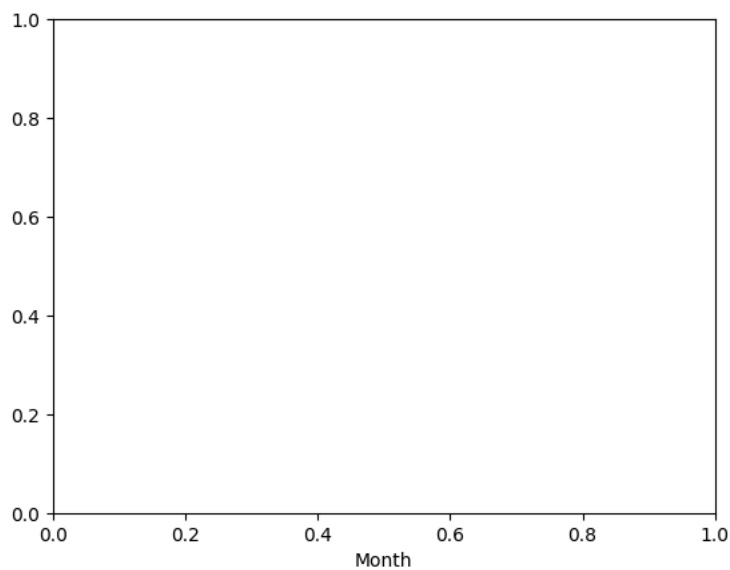


```
#used to provide title for this graph to understand purpose of graph  
plt.title('Monthly Sales vs Costs', fontsize=16, fontweight='bold')
```

Show hidden output

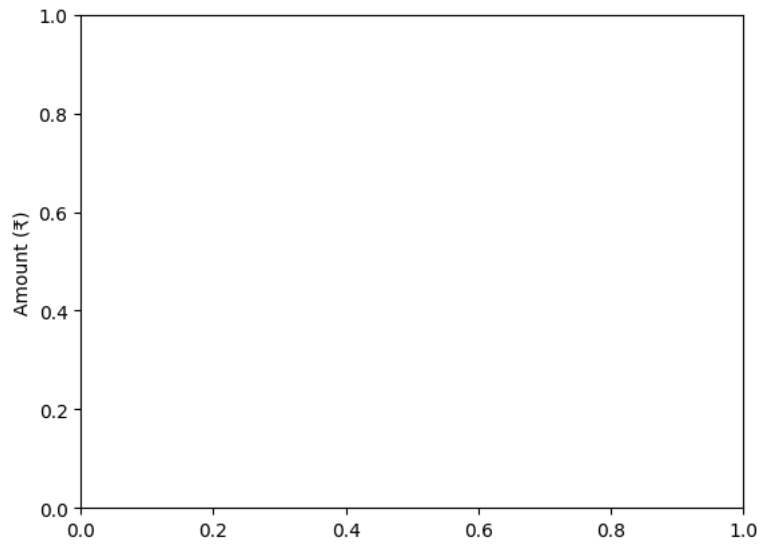
```
#adds label to x-axis  
plt.xlabel('Month')
```

```
Text(0.5, 0, 'Month')
```



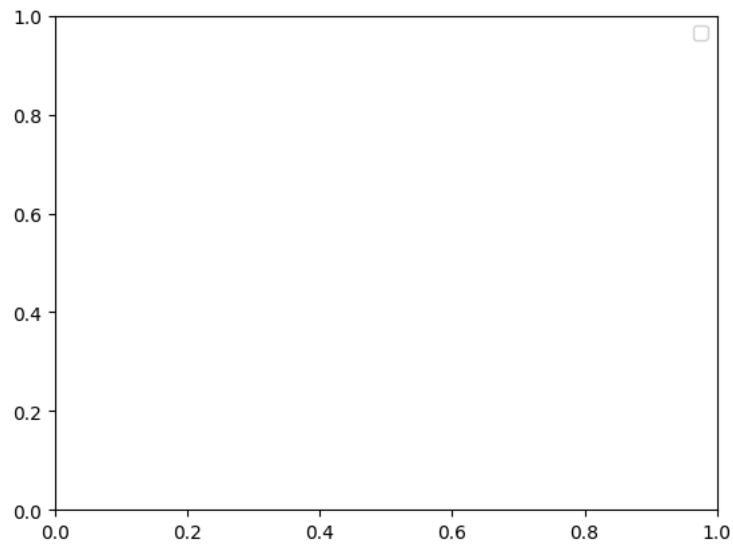
```
#adds lable to y-axis  
plt.ylabel('Amount (₹)')
```

```
Text(0, 0.5, 'Amount (₹)')
```

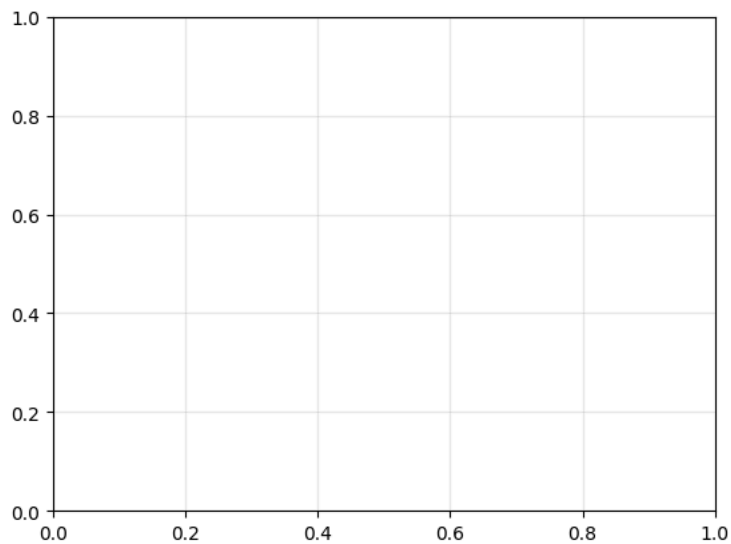


```
# Shows labels for plotted lines.  
plt.legend()
```

```
/tmp/ipykernel_1525/3779859436.py:2: UserWarning: No artists with labels found to put in legend. Note that artists whose labels are not unique are not plotted in legend.  
plt.legend()  
<matplotlib.legend.Legend at 0x7ba3287ab1a0>
```



```
# Adds background grid lines.  
# True=Turns grid ON.  
# alpha= controles tranparency  
plt.grid(True, alpha=0.3)
```



```
#Automatically adjusts spacing.  
plt.tight_layout()
```

<Figure size 640x480 with 0 Axes>

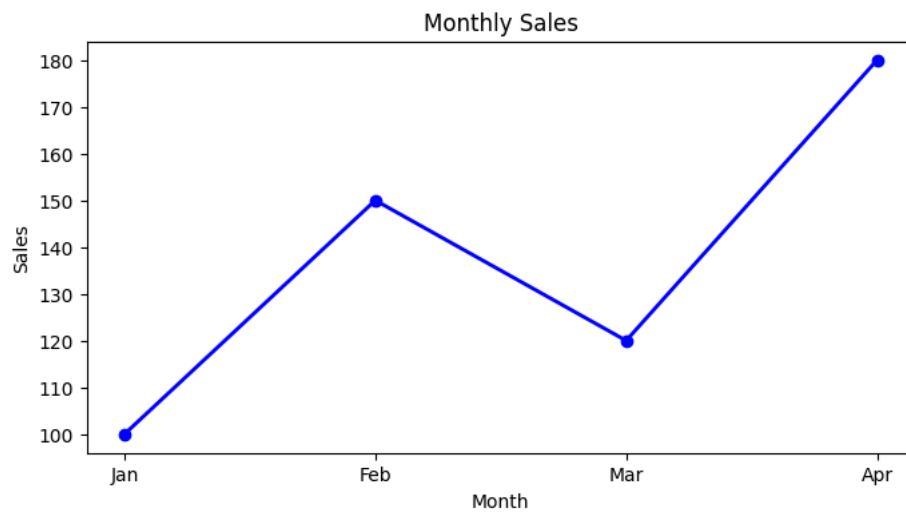
```
# Saves graph as image file.  
# DPI = Dots Per Inch  
plt.savefig('/content/sale_ct.png')
```

<Figure size 640x480 with 0 Axes>

```
plt.show()  
# Displays graph on screen.
```

```
# example code
```

```
import matplotlib.pyplot as plt  
  
months = ['Jan', 'Feb', 'Mar', 'Apr']  
sales = [100, 150, 120, 180]  
  
# Create figure  
plt.figure(figsize=(8,4))  
  
# Plot data  
plt.plot(months, sales,  
         marker='o',  
         color='blue',  
         linewidth=2)  
  
# Labels and title  
plt.title("Monthly Sales")  
plt.xlabel("Month")  
plt.ylabel("Sales")  
  
# Save FIRST  
plt.savefig('/content/sales_chart.png', dpi=300)  
  
# THEN show  
plt.show()
```



bar chart

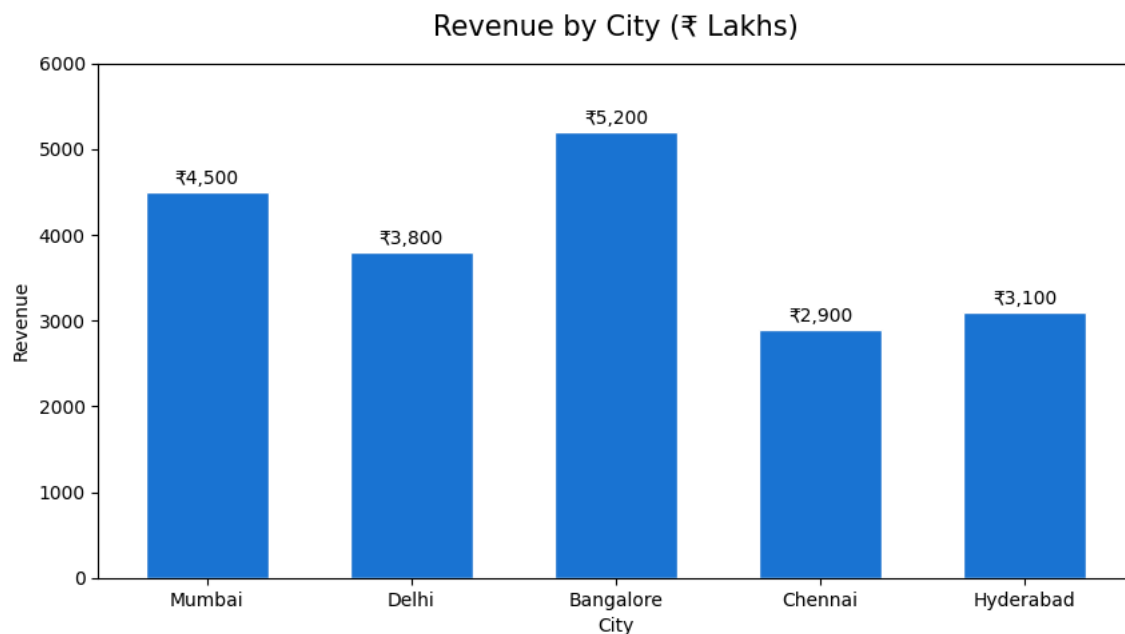
```
cities = ['Mumbai', 'Delhi', 'Bangalore', 'Chennai', 'Hyderabad']
revenue = [4500, 3800, 5200, 2900, 3100]

fig, ax = plt.subplots(figsize=(10, 5)) # OOP style (preferred)

bars = ax.bar(cities, revenue, color='#1976D2', edgecolor='white', width=0.6)

# Add value labels ON TOP of each bar
for bar in bars:
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2, height + 50,
            f'₹{height:,.0f}', ha='center', va='bottom', fontsize=10)

ax.set_title('Revenue by City (₹ Lakhs)', fontsize=15, pad=15)
ax.set_xlabel('City')
ax.set_ylabel('Revenue')
ax.set_ylim(0, 6000) # set y-axis range
plt.savefig('revenue_bar.png', dpi=150, bbox_inches='tight')
plt.show()
```



histogram

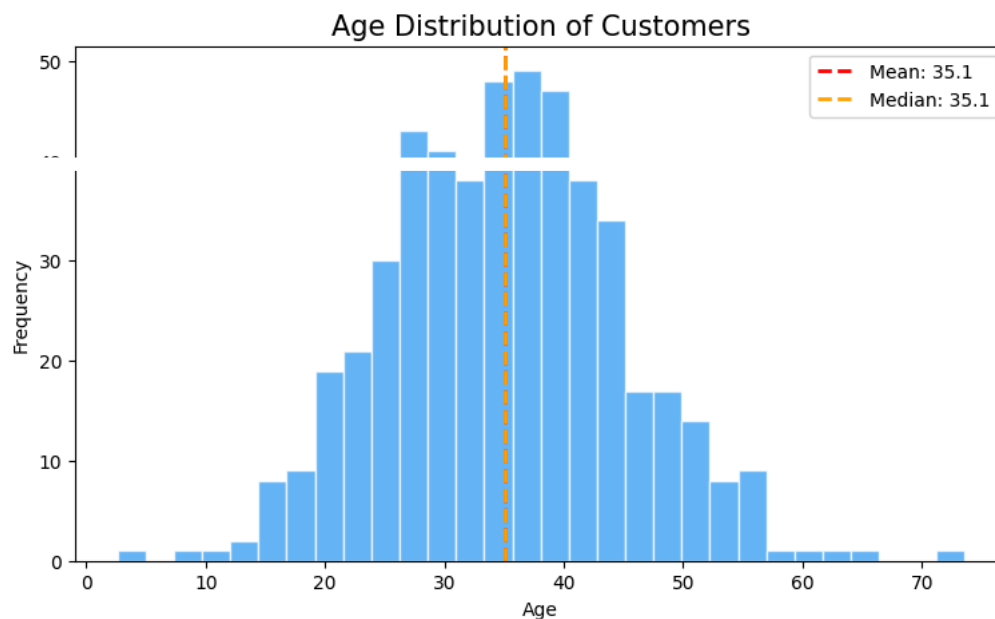
```
np.random.seed(42)
ages = np.random.normal(35, 10, 500) # 500 ages, mean=35, std=10

plt.figure(figsize=(9, 5))
plt.hist(ages, bins=30, color='#42A5F5', edgecolor='white', alpha=0.8)

# Add vertical mean line
```

```
plt.axvline(ages.mean(), color='red', linestyle='--', linewidth=2, label=f'Mean: {ages.mean():.1f}')
plt.axvline(np.median(ages), color='orange', linestyle='--', linewidth=2, label=f'Median: {np.median(ages):.1f}')

plt.title('Age Distribution of Customers', fontsize=15)
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.legend()
plt.savefig('age_hist.png', dpi=150)
plt.show()
```



Scatter Plot — Relationships

```
np.random.seed(0)
hours_studied = np.random.uniform(1, 10, 100)
exam_score = 40 + 5 * hours_studied + np.random.normal(0, 5, 100)
study_group = np.random.choice(['A', 'B', 'C'], 100)

colors = {'A': '#1976D2', 'B': '#E53935', 'C': '#43A047'}
color_list = [colors[g] for g in study_group]

plt.figure(figsize=(9, 6))
scatter = plt.scatter(hours_studied, exam_score,
                      c=color_list, alpha=0.7, s=60, # s = marker size
                      edgecolors='white', linewidth=0.5)

# Legend for groups
for group, col in colors.items():
    plt.scatter([], [], c=col, label=f'Group {group}', s=60)

plt.title('Hours Studied vs Exam Score', fontsize=15)
plt.xlabel('Hours Studied')
plt.ylabel('Exam Score')
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig('study_scatter.png', dpi=150, bbox_inches='tight')
plt.show()
```

